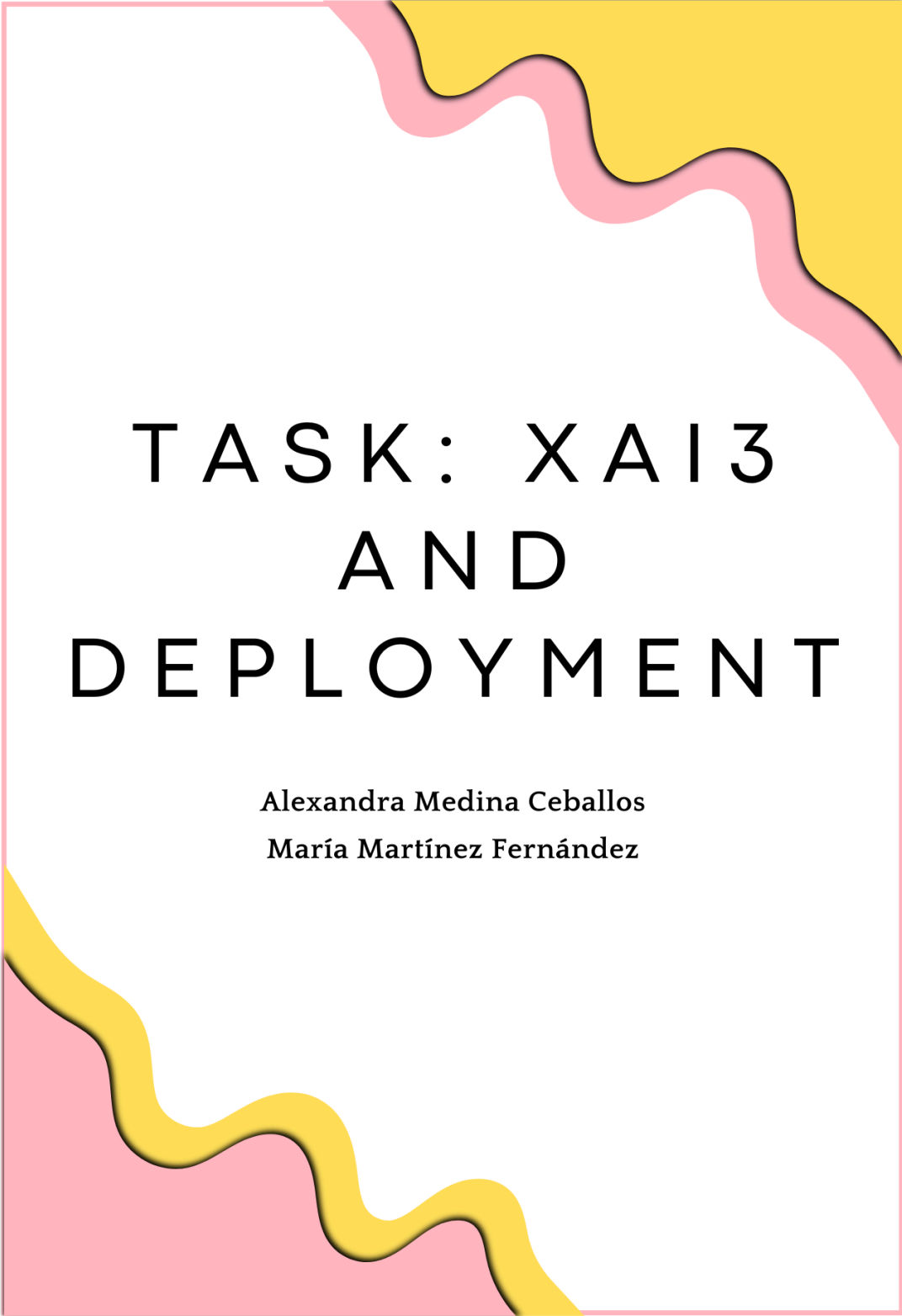




TASK: XAI3 AND DEPLOYMENT

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1. Introduction

Partial Dependence Plots (PDP) are one of the most widely used model-agnostic interpretability techniques in the field of Explainable AI (XAI). Introduced by Friedman (2001) in the context of gradient boosting machines, they have since become a standard diagnostic tool across all families of supervised learning models. The core idea is to show the marginal effect of one or two features on the predicted outcome of a trained model, after integrating out (averaging over) the influence of all remaining features.

In this practice, Random Forest regression models were trained on two real-world datasets and their internal learned relationships were visualised using PDPs. The datasets used are:

- **Bike rental dataset (*day.csv*)**. Predicting daily bicycle rental counts based on weather and calendar features from the Capital Bikeshare system (Washington D.C., 2011–2012).
- **House price dataset (*kc_house_data.csv*)**. Predicting residential sale price based on structural property characteristics from King County, Washington State.

The full implementation was carried out in R using the `randomForest`, `pdp`, and `ggplot2` libraries. One dimensional PDPs were generated for individual features of interest; a two-dimensional PDP was produced for the joint effect of temperature and humidity; and density plots were added to contextualise predictions with the actual distribution of observed data.

Unlike traditional feature importance measures, PDPs provide information not only about the relative influence of variables, but also about the functional shape, direction, and magnitude of their relationship with model predictions.

2. One-Dimensional Partial Dependence Plots

The Capital Bikeshare dataset contains daily bicycle rental records combined with meteorological and calendar information from 2011 and 2012. Predicting bike demand is a common regression problem in urban mobility analytics, since rental activity is strongly influenced by weather conditions.

The dataset was preprocessed by encoding categorical variables, converting meteorological features back to physical units, and creating a temporal variable called `days_since_2011` to capture the growth of the service over time.

A Random Forest model with 100 trees was trained using the complete feature set to predict the total number of daily rentals (`cnt`). Partial Dependence Plots (PDPs) were then used to analyze the marginal effects of four key variables: `days_since_2011`, temperature, humidity, and wind speed.

2.1 Days Since 2011

The temporal variable `days_since_2011` encodes the number of days elapsed since January 1st, 2011: the start of the service's operation. Its PDP captures the structural growth of bike-sharing demand over time, independently of weather fluctuations.

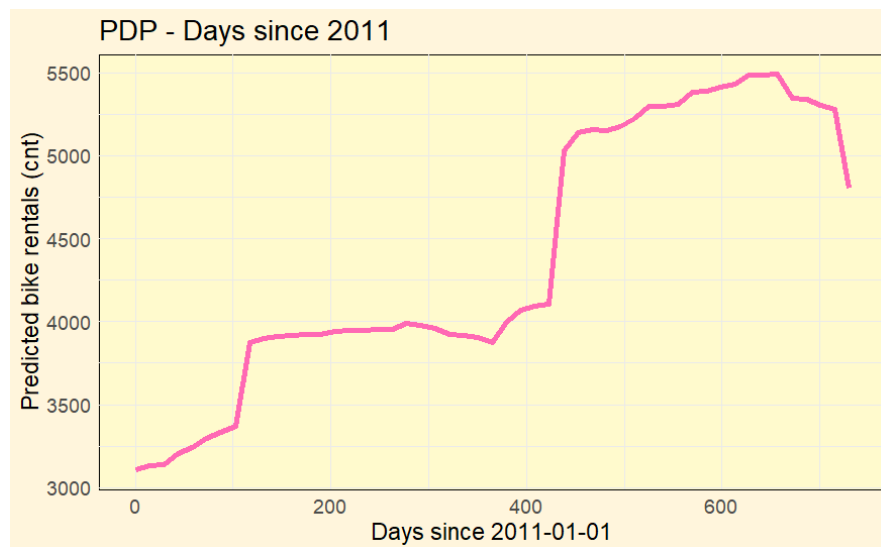


Figure 1. PDP – Days since January 1st, 2011. Predicted daily rentals as a function of elapsed days.

The PDP reveals a stepped rather than smooth growth pattern. Predicted rentals rise sharply in the first approximately 100 days (from ~3,100 to ~3,900), reflecting the initial rapid uptake of the service in Spring 2011. The curve then plateaus until around day 350, followed by a second pronounced increase between days 400 and 500 (reaching ~5,100). It stabilises again until day 650, then shows a slight decline toward the end of the observation period. This stepped structure likely reflects the interaction between two annual seasonal cycles: each spring and summer brings a surge in demand, and each autumn and winter brings a contraction. Crucially, the overall upward displacement between the two years confirms genuine market growth: the service gained users across both seasonal troughs and peaks over the observation period.

2.2 Temperature

Temperature is one of the most theoretically motivated predictors of outdoor physical activity. Cold temperatures increase physical effort and discomfort; extreme heat can also discourage exercise. The PDP isolates the marginal effect of temperature while averaging over all other conditions.

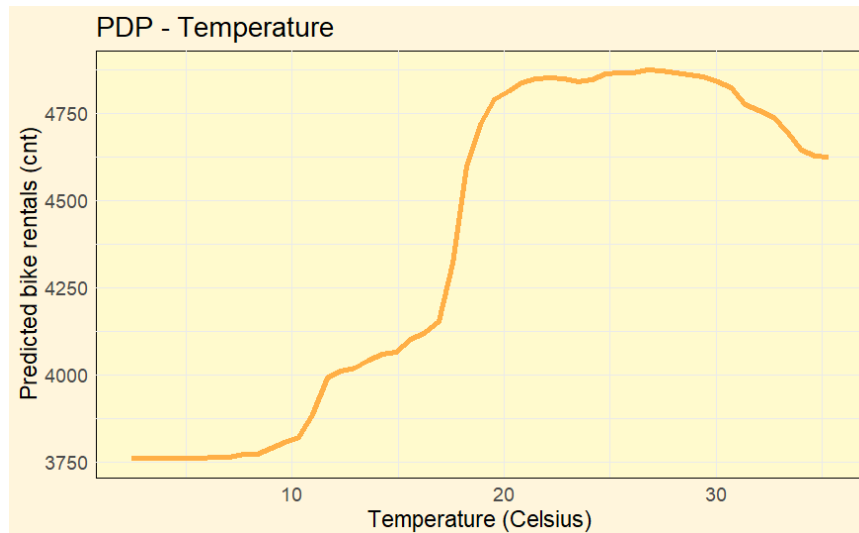


Figure 2. PDP – Temperature (°C). Marginal effect of temperature on predicted daily bike rentals.

The relationship follows a clear S-shaped (sigmoidal) curve. Predictions remain low and flat below 8°C (approximately 3,750 rentals per day), indicating that cold weather strongly suppresses demand regardless of other conditions. Between 10°C and 20°C, predictions rise steeply: this captures the transition from cold to comfortable cycling conditions. The curve stabilises between 20°C and 30°C at around 4,850 rentals: the thermal comfort zone where weather ceases to be a limiting factor. Above 30°C, predictions noticeably decline to approximately 4,600, suggesting that extreme heat also discourages cycling. Temperature is the single most influential predictor identified in this exercise, with a prediction range exceeding 1,100 rentals across its observed domain.

2.3 Humidity

Unlike temperature, humidity does not affect cycling through a simple monotonic mechanism. The PDP reveals an important threshold effect that a linear model would be unable to capture.

The density plots also highlight one of the main methodological limitations of Partial Dependence Plots. PDPs assume partial independence between the analysed feature and the remaining predictors. When strong correlations exist between variables, the averaging process may generate unrealistic or sparsely represented feature combinations. Consequently, predictions in low-density regions should be interpreted cautiously, as they may correspond to extrapolations beyond the empirical distribution of the training data rather than reliable interpolations.

This limitation is particularly relevant for the outer regions of the 2D PDP, especially combinations involving very low humidity or extreme temperatures, where the number of observed training instances is considerably smaller. In these regions, the Random Forest model has limited evidence available to support stable predictions.

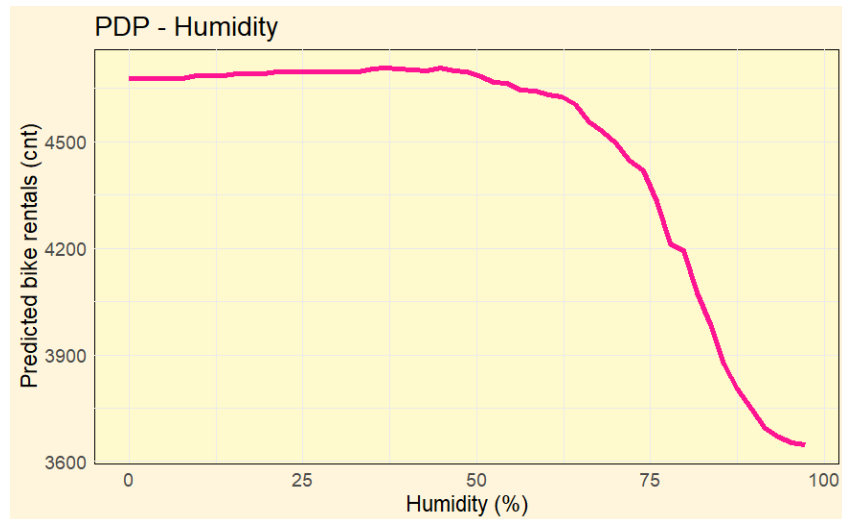


Figure 3. PDP – Humidity (%). Marginal effect of relative humidity on predicted daily bike rentals.

2.4 Wind Speed

Wind speed influences both the physical difficulty of cycling (aerodynamic resistance, balance) and perceived comfort. The PDP reveals a combination of a gradual and an abrupt component in this relationship.

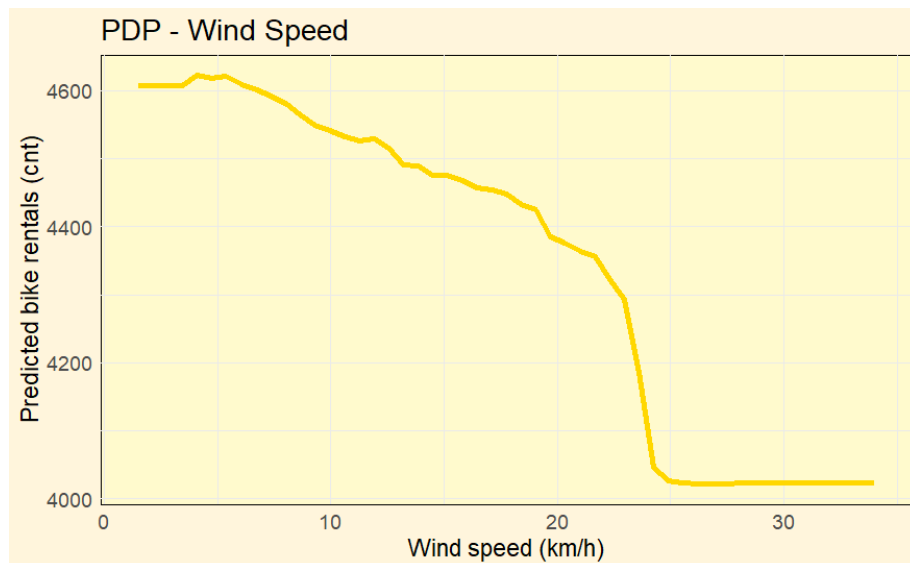


Figure 4. PDP – Wind Speed (km/h). Marginal effect of wind speed on predicted daily bike rentals.

The PDP shows a gradual decrease in predicted rentals as wind speed increases from 0 to approximately 20 km/h (from ~4,500 to ~4,350), consistent with progressively increasing resistance. Around 25 km/h, however, there is a very abrupt drop (from ~4,350 to ~4,050), after which the curve flattens completely. This saturation effect suggests that beyond a certain wind intensity, additional wind no longer changes rider behaviour, users deterred already have stopped, and those who remain are unaffected by further increases.

The threshold at ~25 km/h is the most notable feature of this PDP and may correspond to a recognized comfort boundary in outdoor human activity research.

3. Two-dimensional Partial Dependence Plot

One-dimensional PDPs treat features independently, but real-world phenomena are often characterised by interaction effects: the joint influence of two variables may differ substantially from the sum of their individual marginal effects. Two-dimensional PDPs extend the concept by computing the average predicted outcome across a full grid of combinations of two features, enabling the visualisation of joint effects and potential synergies or dampening interactions.

Due to the computational cost of generating a 2D PDP, which requires evaluating the model across a two dimensional grid for each training observation, a random subsample of 300 rows was extracted from the preprocessed bike dataset before generating the partial dependence surface. A grid resolution of 20 was used for both axes, producing a 20×20 matrix of average predictions visualised as a heatmap using `geom_tile()`.

However, interpreting 2D PDPs requires additional caution, since sparse combinations of feature values may produce unstable estimations due to limited representation within the training data distribution.

3.1 2D PDP – Temperature & Humidity

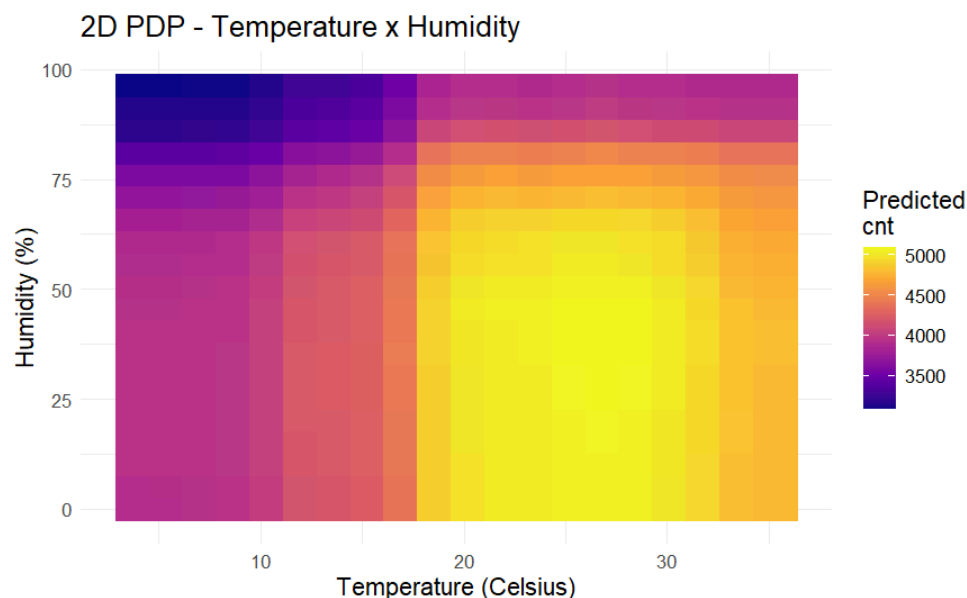


Figure 5. 2D PDP – Temperature × Humidity. Each tile shows the average predicted daily rental count for that feature combination. Yellow = high demand; purple = low demand.

The 2D PDP reveals the joint marginal effect of temperature and humidity on predicted bike rentals. Warmer colours (yellow and orange) indicate higher predicted counts; cooler colours (purple and blue) indicate lower predictions. The most striking feature of the plot is that the colour pattern is predominantly vertical, within each temperature column, the colour remains

largely consistent across all humidity levels. This confirms that temperature is by far the dominant driver of the joint prediction.

The highest predicted rental values appear in the high-temperature region (above 20°C) combined with low-to moderate humidity (below 75–80%). When humidity rises above 80%, even at high temperatures, colours shift toward orange-pink, confirming that extreme humidity reduces predictions, but only at that upper extreme. At low temperatures (below 10°C), predictions are consistently low regardless of humidity level, confirming that cold weather is a strong deterrent that no amount of humidity variation can compensate for.

3.2 Density distributions

Density plots of the input features are essential companions to the 2D PDP, as they reveal which regions of the feature space are well-represented in the training data. Predictions in sparse regions are extrapolations rather than interpolations, and should be interpreted with considerable caution.

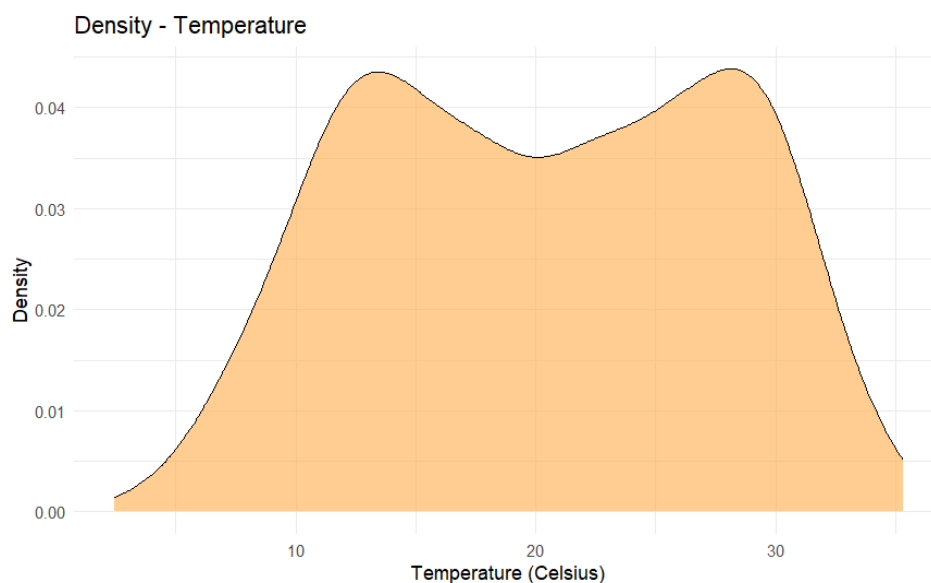


Figure 6. Density – Temperature (°C).

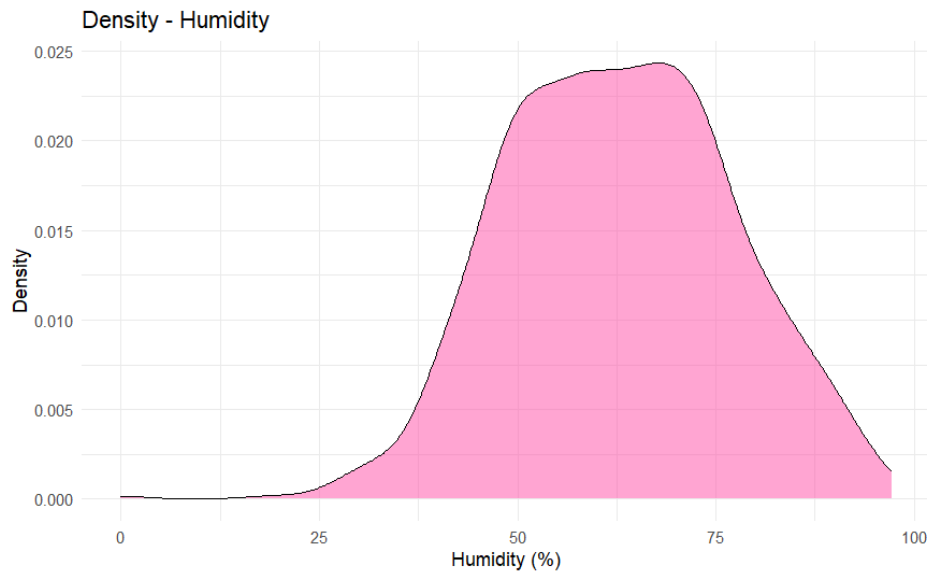


Figure 7. Density – Humidity (%).

Temperature follows a bimodal distribution with peaks around 14°C and 28°C, reflecting the two annual seasonal cycles across 2011–2012. Humidity is concentrated between 50% and 80%, with very few observations below 25%. This means the bottom-left corner of the 2D plot, low humidity at any temperature, represents a sparse region of the training data, and the predicted values there are unreliable extrapolations.

Temperature dominates the joint prediction surface; the colour gradient is predominantly vertical, meaning changes in temperature drive most of the variation in predicted rentals. Humidity acts as a secondary moderator, only visibly dampening rentals above ~80%. The density plots confirm that the central zone of the 2D plot (10–32°C, 45–80% humidity) contains the bulk of training data and therefore the most reliable predictions. Sparse outer regions, particularly low humidity and extreme temperatures, should be interpreted cautiously as they represent areas with limited evidence in the training data.

4. PDP for House Price Prediction

The King County dataset contains sale records for over 21,000 residential properties in the greater Seattle metropolitan area, covering structural characteristics (bedrooms, bathrooms, living area, lot area, floors, and year built), condition and quality ratings, location features, and sale prices for transactions occurring between 2014 and 2015. House price prediction is both a canonical machine learning benchmark and a problem of direct economic and social relevance: accurate property valuation affects real-estate markets, mortgage lending, investment decisions, and urban planning policy.

For this exercise, a Random Forest with 100 trees was trained on a stratified random subsample of 3,000 rows to manage computational cost while preserving the distributional properties of the full dataset. The model uses six features: bedrooms, bathrooms, sqft_living, sqft_lot, floors, and yr_built. PDPs were then generated for the four variables of interest using the same subsample to ensure consistency.

4.1 Bedrooms

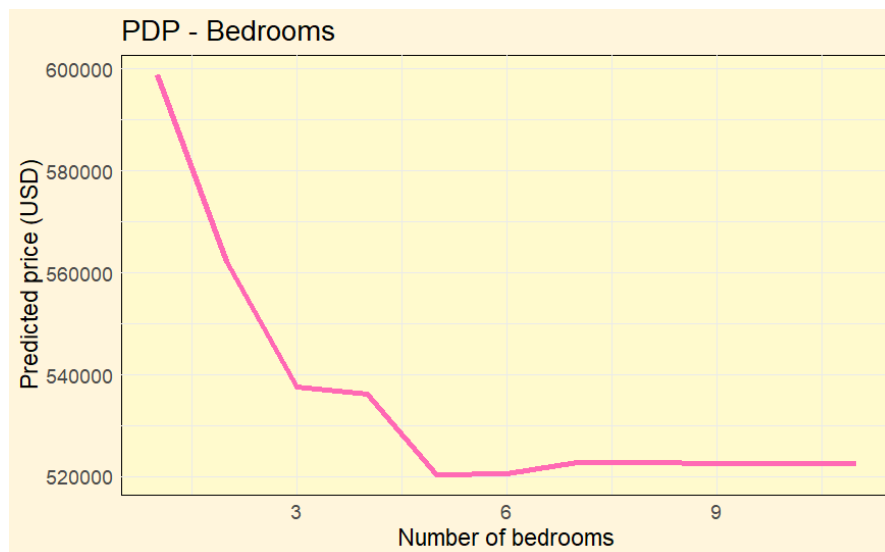


Figure 8. PDP – Bedrooms. Marginal effect of bedroom count on predicted house sale price (USD)

The PDP for bedrooms reveals a counterintuitive negative relationship: properties with 1–2 bedrooms are associated with the highest predicted prices (~\$600,000), which drop sharply to around \$538,000 at 3 bedrooms and continue declining to approximately \$520,000 at 5–6 bedrooms, after which the curve flattens. This result appears paradoxical but has a clear economic interpretation. The PDP is a conditional marginal effect: it shows the effect of adding bedrooms while holding all other features, including `sqft_living`, constant.

A fixed-size property with more bedrooms necessarily has smaller individual rooms, which buyers in this market appear to perceive negatively. This reflects a classic real-estate trade-off between room count and room size: in isolation, more bedrooms do not add value; in the context of a fixed living area, they can actively reduce it. Bedrooms is therefore not a straightforward value driver when square footage is already in the model.

This interpretation also illustrates an important characteristic of PDPs: the plots represent conditional marginal effects rather than isolated causal relationships. The negative effect observed for bedrooms does not imply that adding bedrooms universally decreases property value. Instead, it reflects the effect of increasing bedroom count while holding total living area constant, effectively reducing average room size within the property.

4.2 Bathrooms

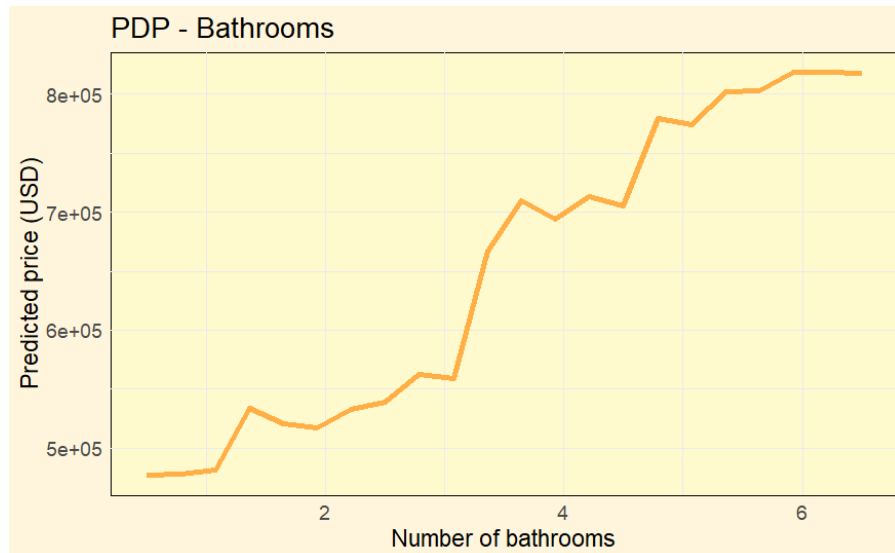


Figure 9. PDP – Bathrooms. Marginal effect of bathroom count on predicted house sale price (USD).

Bathrooms show a clear and strong positive relationship with predicted price. The curve rises from approximately \$470,000 at 1 bathroom to over \$820,000 at 6–7 bathrooms, with the steepest gains occurring between 2 and 4 bathrooms. The relationship is not perfectly smooth, there are minor local dips at certain values, but the overall upward trend is robust and consistent across the full range.

Unlike bedrooms, bathrooms add measurable value even when all other structural features are held constant. This likely reflects two mechanisms: bathrooms are expensive to construct (requiring plumbing, fixtures, and waterproofing), so their presence signals higher construction quality and investment in the property; and additional bathrooms directly improve household functionality and comfort, particularly for larger families or in the context of luxury properties, in a way that additional bedrooms in a fixed-size space simply do not.

4.3 Living Area

Living area is by far the most influential predictor in the model, both in terms of the range of values it spans and the consistency and magnitude of its effect. Predicted price rises nearly monotonically from approximately \$330,000 at 500 sq ft to over \$1,500,000 at 8,000 sq ft, a total range exceeding \$1,200,000. The increase is steepest between 1,000 and 5,000 sq ft, which is precisely where the majority of residential properties in the market fall.

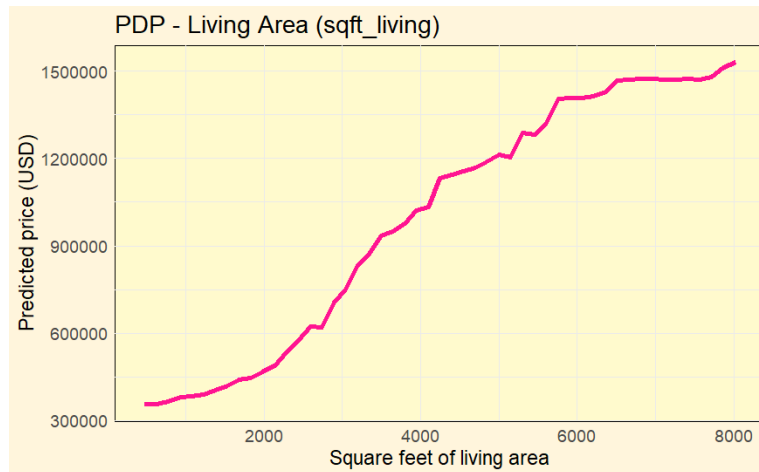


Figure 10. PDP – Living Area (*sqft_living*). Marginal effect of interior living space on predicted house sale price (USD).

The dominance of *sqft_living* is entirely consistent with economic theory: interior space is scarce, costly to construct and maintain, and buyers are willing to pay a strong premium for it. The relationship is roughly linear through the most data-dense region, becoming steeper at the upper end as luxury properties command disproportionate premiums per additional square foot. This single variable alone explains the vast majority of predicted price variation in the model: its prediction range is more than ten times larger than that of any other feature examined.

4.4 Floors

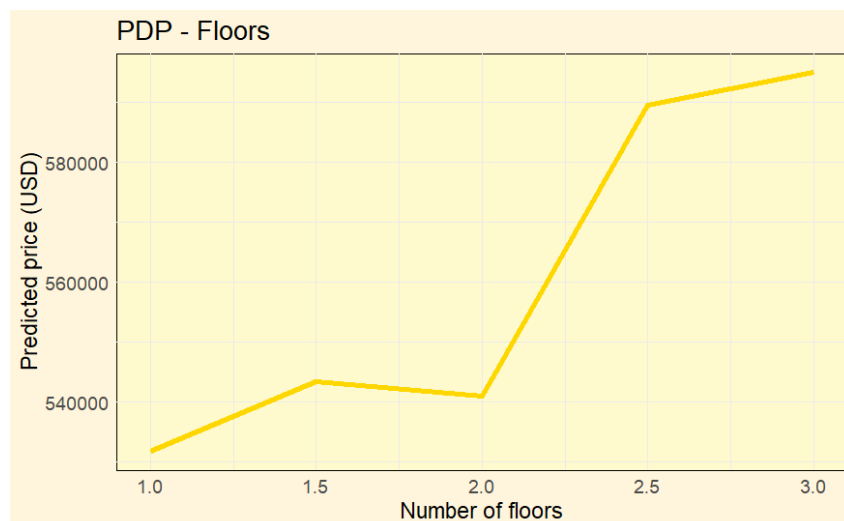


Figure 11. PDP – Floors. Marginal effect of floor count on predicted house sale price (USD).

The effect of floors is modest relative to the other variables examined. Predicted prices increase slightly from 1 to 1.5 floors (\$530,000 to \$543,000) and remain nearly flat up to 2 floors (\$541,000). The most notable jump occurs between 2 and 2.5 floors, where predictions rise sharply from \$541,000 to approximately \$587,000. Above 2.5 floors, the curve continues to rise slightly to \$595,000 at 3 floors.

The overall prediction range is approximately \$65,000, making floors the weakest predictor among the four examined. The step-like pattern reflects that floor count in this dataset takes discrete values (1, 1.5, 2, 2.5, 3), with the main value-adding threshold lying between a standard two-storey property and a split-level or multi-storey configuration. Properties with more floors tend to co-occur with other high-value characteristics not fully captured by the six included features, which may explain the premium at higher floor counts.

sqft_living dominates the prediction with a price range exceeding \$1.2M, it is the overwhelmingly primary driver of house value and dwarfs all other variables. Bathrooms are a strong secondary predictor with a consistent positive effect reflecting construction quality and household functionality signals. Floors contribute positively but modestly, with the main value premium concentrated above the 2-storey threshold. Bedrooms produce a counterintuitive negative marginal effect: once living area is held constant, more bedrooms signal smaller rooms and can actively reduce predicted price.

5. Conclusions

This practice demonstrated that Partial Dependence Plots are a powerful and interpretable tool for understanding Random Forest models, going beyond standard feature importance by revealing the shape, direction, and magnitude of each feature's relationship with the prediction.

For bike rental prediction, temperature was the main driver of demand, with rentals peaking between 20–30°C and decreasing under extreme temperatures. Humidity and wind speed showed clear non-linear threshold effects, while the variable *days_since_2011* captured the structural growth of the bike-sharing service over time. The 2D PDP also confirmed the dominant role of temperature, with humidity acting mainly as a secondary moderator.

For house price prediction, living area (*sqft_living*) was by far the strongest predictor, followed by bathrooms and number of floors. Bedrooms showed a counterintuitive negative marginal effect once living area was controlled for, reflecting the trade-off between room size and room count.

Although PDPs provide highly intuitive visual explanations of model behaviour, they also present important limitations that must be considered during interpretation. Since PDPs display average marginal effects across the dataset, they may hide local heterogeneity and instance-specific responses. Furthermore, the method does not provide causal explanations, only statistical relationships learned by the predictive model.

Nevertheless, despite these limitations, PDPs remain one of the most effective and accessible explainability techniques for analysing complex non-linear machine learning models. Their ability to translate high-dimensional predictive behaviour into interpretable visual patterns makes them particularly valuable in applied machine learning and decision-support systems.

5.1 Business Deployment

The insights extracted through PDPs have direct business applications beyond academic analysis. For mobility operators, understanding how weather affects bike demand can improve fleet allocation, maintenance planning, and dynamic pricing strategies. For real-estate companies and investors, PDPs provide a clear hierarchy of value drivers that can support pricing, renovation, and investment decisions.

More broadly, PDPs contribute to the growing need for explainable AI in regulated industries. They provide transparent and communicable visualisations that help stakeholders understand and validate model behaviour without requiring detailed knowledge of the underlying algorithms.

Explainability techniques such as PDPs transform black-box models into auditable and actionable decision-support tools. Beyond improving interpretability, they support transparency, governance, and responsible AI deployment in real-world environments.

Beyond operational and business applications, explainability techniques such as PDPs also contribute to the broader objectives of Responsible AI. Transparent model interpretation improves auditability, facilitates regulatory compliance, and strengthens stakeholder trust in automated decision-making systems.

In practical environments, explainable AI techniques help bridge the gap between highly complex predictive models and human decision makers. By providing interpretable visual evidence of how predictions are generated, PDPs support more transparent, accountable, and trustworthy AI deployment across both technical and non-technical domains.

Explainability techniques such as PDPs transform black-box models into auditable and actionable decision-support tools. Beyond improving interpretability, they support transparency, governance, and responsible AI deployment in real-world environments. As machine learning systems continue to expand into increasingly sensitive domains, the integration of explainable AI methodologies will become not only desirable, but essential for trustworthy and operationally reliable artificial intelligence systems.